Earthlike planet on which the surface became hot enough to melt tin or lead, is worth considering—especially by those who say that the increasing greenhouse effect on Earth will be self-correcting, that we don't really have to worry about it, or (you can see this in the publications of some groups that call themselves conservative) that the greenhouse effect itself is a "hoax."

(3) Nuclear winter is the predicted darkening and cooling of the Earth—mainly from fine smoke particles injected into the atmosphere from the burning of cities and petroleum facilities—that is predicted to follow a global thermonuclear war. A vigorous scientific debate ensued on just how serious nuclear winter might be. The various opinions have now converged. All three-dimensional general circulation computer models predict that the global temperatures resulting from a worldwide thermonuclear war would be colder than those in the Pleistocene ice ages. The implications for our planetary civilization—especially through the collapse of agriculture—are very dire. It is a consequence of nuclear war that was somehow overlooked by the civil and military authorities of the United States, the Soviet Union, Britain, France, and China when they decided to accumulate well over 60,000 nuclear weapons. Although it's hard to be certain about such things, a case can be made that nuclear winter played a constructive role (there were other causes, of course) in convincing the nuclear-armed nations, especially the Soviet Union, of the futility of nuclear war.

Nuclear winter was first calculated and named in 1982/83 by a group of five scientists, to which I'm proud to belong. This team was given the acronym TTAPS (for Richard P. Turco, Owen B. Toon, Thomas Ackerman, James Pollack, and myself). Of the five TTAPS scientists, two were planetary scientists, and the other three had published many papers in planetary science. The earliest intimation of nuclear winter came during that same *Mariner 9* mission to Mars, when there was a global dust storm and we were unable to see the surface of the planet; the infrared spectrometer on the spacecraft found the high atmosphere to be warmer and the surface colder than they ought to have been. Jim Pollack and I sat down and tried to calculate how that could come about. Over the subsequent twelve